MILITARY SPECIFICATION

HOSE ASSEMBLY OUTFIT (REUSABLE ADAPTERS)

This specification is approved for use by the U.S. Army Armament, Research and Development Center, Department of the Army and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a hydraulic hose assembly outfit and includes hoses, industry universal and reusable hose adapters, steel tubing adapters and nipples, hose cut-off machine, hose assembling machine and associated equipment utilized in the fabrication and repair or replacement of hydraulic hose assemblies.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issue of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

PPP-B-601	-	Boxes, Wood, Cleated-Plywood
PPP-B-621	-	Boxes, Wood, Nailed and Lock-Corner
PPP-B-636	-	Boxes, Shipping, Fiberboard
PPP-B-640	-	Boxes, Fiberboard, Corrugated, Triple-Wall
PPP-F-320	-	Fiberboard: Corrugated and Solid, Sheet Stock (Container Grade) and Cut shapes
PPP-T-60	-	Tape: Packaging, Waterproof

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research, Development and Engineering Center, ATTN: SMCAR-EST-S, Rock Island, IL 61299-7300 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MILITARY

MIL-P-116	-	Preservation, Methods of
MIL-B-121	-	Barrier Material, Greaseproofed, Waterproofed, Flexible
MIL-P-775	-	Packaging of Hose, Hose Assemblies; Rubber Plastic, Fabric, or Metal (including tubing), and Fittings, Nozzles, and Strainers
MIL-C-5501	-	Caps and Plugs, Protective, Dust and Moisture Seal
MIL-H-6083	-	Hydraulic Fluid, Petroleum Base, Preservation and Operation
MIL-F-18866	-	Fitting, Hydraulic Tube, Flared, 37 Degree & Flareless Steel
MIL-H-52471	-	Hose & Hose Assemblies, Rubber, Hydraulic Pressure Type, General Specification For
MIL-H-52471/5	-	Hose, Rubber, Hydraulic Pressure Type, Type 100R2, Double Wire Braid Reinforcement Hose
MIL-H-52471/6	-	Hose, Rubber, Hydraulic Pressure Type, Type 100R10, Four Spiral Wrap Reinforcement Hose
MIL-F-52525	-	Fitting, Wire Reinforced Hydraulic Hose & Clamp Halves, General Specification for
MIL-F-52525/1	-	Fitting, Wire Reinforced Hydraulic Hose & Clamp Halves, Field Attachable (reusable), Screw-on to 37 Degree Flare Female Swivel, Straight, Type B, For Use With Type 100R2 Double Wire Braid Hose
MIL-F-52525/9	-	Fitting, Wire Reinforced Hydraulic Hose & Clamp Halves, Field Attachable (reusable), Screw-on to 37 Degree Flare Female Swivel, Straight, Type C, For Use With Type 100R10 4 Spiral Wrap Hose
MIL-F-52525/16	-	Fitting, Wire Reinforced Hydraulic Hose & Clamp Halves, Clamp Halves For 4 Bolt Split Flange Fitting
MIL-F-52525/23	-	Fittings, Wire Reinforced Hydraulic Hose & Clamp Halves, Field Attachable (reusable), Clamp-on to 4 Bolt Split Flange, 45 Degree Bent Tube, Type C, Adapter, For Use With 100R10 4 Spiral Wrap Hose
MIL-F-52525/24	-	Fittings, Wire Reinforced Hydraulic Hose & Clamp Halves, Field Attachable (reusable), Clamp-on to 4 Bolt Split Flange, 45 Degree Bent Tube, Type C, Elbow, For Use with Type 100R10 4 Spiral Wrap Hose
MIL-F-52525/25	-	Fittings, Wire Reinforced Hydraulic Hose & Clamp Halves, Field Attachable (reusable), Clamp-on to 4 Bolt Split Flange, 90 Degree Bent Tube, Short Drop, Type C Elbow, For Use With Type 100R10 4 Spiral Wrap Hose

STANDARDS

FEDERAL

FED-STD-H28	-	Screw-Thread Standards for Federal Services

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking For Shipment and Storage

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MIL-STD-130	-	Identification Marking of U.S. Military Property
MIL-STD-889	-	Dissimilar Metals
MIL-STD-1186	-	Cushioning, Anchoring, Bracing, Blocking and Waterproofing; with appropriate Test Methods
MIL-STD-1190	-	Minimum Guidelines for Level C Preservation, Packing and Marking
MS51500	-	Adapter, Straight, Pipe to Tube, Male Pipe End, 37 Degrees Flared
MS51501	-	Nipple, Tube, Regular Hex, 37 Degrees Flared
MS51525	-	Adapter, Straight, Tube To Boss, O-Ring, 37 Degrees Flared

(Unless otherwise indicated, copies of federal and military specifications and standard are available from the Naval Publications and Forms Center, (ATTN: NPDOS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government documents & drawings. The following other Government documents and drawings form a part of this document to the extent specified herein. Unless otherwise specified, the issues shall be those cited in the solicitation.

DRAWINGS

FORT BELVOIR

D13221E6701 - Power Saw Assembly, Hose Cutter

(Unless otherwise indicated, copies of drawings are available from the U.S. Army Belvoir Research, Development and Engineering Center, Fort Belvoir, VA 22060-5606.)

PUBLICATIONS

U.S. DEPARTMENT OF LABOR

29 CFR, Chapter XVII, Part 1910 - Occupational Safety and Health Standards

(Unless otherwise indicated, copies of Code of Federal Regulations (CFR) are available from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issue of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIAL (ASTM)

ASTM B117 - Salt Spray (Fog) Testing, Method of

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103)

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CATERPILLAR TRACTOR

Part Number - 7S1126
Part Number - 7S1120

(Applications for copies should be addressed to Caterpillar Tractor, 100 N.E. Adams St., Peoria, IL 61629.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE J525 - Welded and Cold Drawn Low Carbon Steel Tubing Annealed for Bending and Flaring.

(Applications for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for associated detail specifications sheets or MS standards), the text of this document shall take precedence. Nothing in this document, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.2.1 and 6.2.1).

3.2 Design. Each hose assembly outfit shall be new, and consist of equipment of current design. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, repair, and replacement. The hose assembly outfit shall be utilized to affix industry universal/reusable adapters in sizes from and including 1/4 inch through 2 inches to the hoses specified herein.

3.3 Material. Materials not specifically designated herein or in the contract shall be of a quality commensurate with best commercial practices, shall be suitable for the intended purpose, and shall meet all requirements specified herein. Materials shall be free from defects which would adversely affect the performance or maintainability of the individual components or the overall assembly. When dissimilar metals are used in contact with each other, suitable protection against galvanic corrosion shall be applied in accordance with MIL-STD-889.

3.3.1 Reclaimed materials. The manufacturer may use reclaimed materials for fabricating new parts. Reclaimed materials shall have been reprocessed, remanufactured, or recycled in a manner that restores them to the same chemical composition and physical properties as the materials originally selected for use. Use of reclaimed parts as is or rebuilt from scrap or other used equipment shall not be permitted.

3.4 Construction.

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3.4.1 Castings and forgings. All castings and forgings shall be free from defects which affect serviceability or performance such as fins, scale, inclusions, cold shuts, voids, cracks, thermal ruptures, laps, folds, mismatching, etc. Defective castings and forgings shall not be reclaimed for use on the hose assembly outfit.

3.4.2 Welding, brazing and soldering. Welding, brazing and soldering shall be of a quality which shall sustain all requirements of the welded, brazed or soldered parts. Welding, brazing, and soldering operations shall not be employed as repair measures for defective parts.

3.4.3 Fastening devices. All screws, pins, bolts, and similar parts shall be installed with means for adjustment and prevention of loss of tightness. All parts subject to removal or adjustment shall not be swaged, peened, staked or otherwise permanently deformed.

3.4.4 Threads. All machined threads shall conform with FED-STD-H28 and the applicable detailed standards referenced therein.

3.4.5 Lubricating systems. All bearings (except sealed for life, permanently lubricated type), matching gears, and all other moving parts shall be provided means to assure adequate lubrication. Recirculating systems shall include filters which are cleanable or replaceable. Each lubricant reservoir shall have means for determining fluid levels. All oil holes, grease fittings, and filler caps shall be so located as to be readily accessible. The supplier shall include and install all lubricants, fluids, greases and other compounds. Vent openings in the lubrication systems shall be temporarily capped or plugged to prevent loss of lubricants during shipment. Tags, warning that "ALL TEMPORARY CAPS OR PLUGS MUST BE REMOVED BEFORE START-UP OR USE OF THE ITEM", shall be attached to all temporary caps or plugs intended to be vented to the atmosphere during operation. These warnings shall also be included in the instruction manual. In the event a lubricant reservoir cannot be covered and sealed, it shall be appropriately preserved and the lubricant shall be provided in a sealed container and shipped with each machine. A warning tag shall be placed in a conspicuous place near the filling area, warning the user to "INSTALL THE FURNISHED LUBRICANT BEFORE OPERATION OF THE EQUIPMENT". The operator's manual shall also reflect the above warning and procedure.

3.4.6 Safety and health requirements. Covers, guards, and other safety devices shall be provided. The safety devices shall not interfere with operation of the hose assembly outfit. The safety devices shall prevent unintentional contact with the guarded part, and shall be removable to facilitate inspection, maintenance and repair of the parts. All machine parts, components, mechanism and assemblies furnished on the hose assembly outfit, whether or not specifically required herein shall comply with the requirements of 29CFR, Chapter XVII, Part 1910 applicable to the hose assembly outfit.

3.4.7 Interchangeability. All parts of the hose assembly outfit shall be manufactured to definite standards and tolerances. All parts of the hose assembly outfit shall be interchangeable without requiring modification of the replaced part or its mating components. All parts subject to possible future replacement shall be assigned a manufacturer's part number and shall be identified in the repair parts manual. All replaceable parts shall provide the same or better form, fit, function, and performance as the parts initially installed in the hose assembly outfit.

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3.5 Performance and product characteristics. The hose assembly outfit shall be capable of being used to affix fittings in table II to the applicable hose in table I, making hose assemblies conforming to MIL-H-52471. The hose assemblies shall show no evidence of defects, damaged or broken components.

3.6 Details of components. Each hose assembly outfit shall consist of the following principal component parts:

a. Hoses
b. Hose adapters
c. Tubing adapters and nipples
d. Hose assembling machine
e. Hose cut-off machines
f. Hose skiver and mandrels
g. Hose oil

3.6.1 Hoses. Hoses furnished with the hose assembly outfit shall be of the type, size and quantity specified in table I and shall be in accordance with MIL-H-52471. Hoses shall be furnished in bulk quantities and marked every 18 inches or less on the layline. Unless otherwise specified (see 6.2.1), bulk hose shall be furnished as follows: 100 ± 10 feet total consisting of one to three pieces, the shortest being at least 15 feet and the next shortest 25 feet; 60 ± 6 feet total, consisting of one or two pieces, the shortest being 15 feet; 20 feet total consisting of one piece; 10 feet total, consisting of one piece.

TABLE I. Wire - reinforced, rubber, hydraulic hose.

Reinforcement type per MIL-H-52471	Size in I.D.	Military specification part number	Length (ft)
100R2	1/4	M52471/5-4	60
100R2	3/8	M52471/5-6	100
100R2	1/2	M52471/5-8	100
100R2	3/4	M52471/5-12	100
100R2	1	M52471/5-16	60
100R10	1	M52471/6-16	20
100R10	1-1/4	M52471/6-20	20
100R10	1-1/2	M52471/6-24	20
100R10	2	M52471/6-32	10

3.6.2 Hose adapters. Hose adapters shall be of the universal and reusable type and shall be of the type, size, and quantity specified in table II. Adapters shall comply with MIL-F-52525, applicable Military Specification Sheet, and be compatible with the hoses specified in table I. The adapters shall be cadmium or zinc plated to a thickness of 0.0002 in. minimum, or phosphate coated (oil finished). The split flange clamp halves (M52525/17-size) shall meet the requirements of a 16 hour salt spray test in accordance with ASTM B117.

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TABLE II. Hydraulic hose adapters

Type per MIL-H-52525	Size		Military specification part number	Quantity (each)
	Hose (in I.D.)	Tube (in O.D.)		
1. Adapter, Hose to tube: reusable screw-on style, straight type, swiveling 37 degree female flare design				
100R2	1/4	1/4	M52525/1-4-4	10
100R2	1/4	5/16	M52525/1-4-5	15
100R2	3/8	3/8	M52525/1-6-6	15
100R2	3/8	1/2	M52525/1-6-8	15
100R2	1/2	1/2	M52525/1-8-8	15
100R2	1/2	5/8	M52525/1-8-10	15
100R2	3/4	3/4	M52525/1-12-12	15
100R2	1	1	M52525/1-15-16	10
100R10	1-1/4	1-1/4	M52525/9-20-20	2
100R10	1-1/2	1-1/2	M52525/9-24-24	2
Alternate acceptable fittings to the two preceding are Adapters, Hose to Tube, 4-bolt Clamp-on style socket, straight 37 degree female flare swivel				
2. Adapter, Hose to Flange: Clamp-on Style, 4-bolt split flange design, for 4 spiral wire wrapped hydraulic hose per MIL-F-52525.				
Straight	1	1	M52525/23-16-16	4
Straight	1-1/4	1-1/4	M52525/23-20-20	4
Straight	1-1/2	1-1/2	M52525/23-24-24	4
Straight	2	2	M52525/23-32-32	2
45 deg. bent	1	1	M52525/24-16-16	2
45 deg. bent	1-1/4	1-1/4	M52525/24-20-20	2
45 deg. bent	1-1/2	1-1/2	M52525/24-24-24	2
45 deg. bent	2	2	M52525/24-32-32	2
		FLANGE (IN)		
90 deg. bent	1	1	M52525/25-16-16	2
90 deg. bent	1-1/4	1-1/4	M52525/25-20-20	2
90 deg. bent	1-1/2	1-1/2	M52525/25-24-24	2
90 deg. bent	2	2	M52525/25-32-32	2
3. Clamp Half: for use with split flange design hydraulic hose adapters				
Clamp Half	1	-	M52525/16-16	2
Clamp Half	1-1/4	-	M52525/16-20	2
Clamp Half	1-1/2	-	M52525/16-24	2
Clamp Half	2	-	M52525/16-32	2

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3.6.3 Tubing adapters and nipples. Tubing adapters and nipples shall be of the type, size and quantity specified in table III and shall comply with MIL-F-18866 and applicable Military Standards (MS).

TABLE III. Adapters and nipples, hydraulic tubing 1/

Type per MIL-H-18866	Size		Military specification part number	Quantity (each)
	Tube (in O.D.)	NPT		
1. Adapter, Tube to Pipe: Steel, straight type, 37 degree male flare and male straight thread on one end; and male NPT thread on the other end per MIL-F-18866.				
Tube to Pipe	1/4	1/4	MS51500B4-4	10
Tube to Pipe	3/8	3/8	MS51500B6-6	10
Tube to Pipe	1/2	1/2	MS51500B8-8	15
Tube to Pipe	3/4	3/4	MS51500B12	10
Tube to Pipe	1	1	MS51500B16	10
Tube to Pipe	1-1/4	1-1/4	MS51500B20	2
Tube to Pipe	1-1/2	1-1/2	MS51500B24	2
2. Adapter, Tube to Boss: Steel, straight type, 37 degree male flare and male straight thread on one end; and "o" ring gasket and male straight thread on the other end per MIL-F-18866.				
Tube to Boss	5/16	-	MS51525B5	5
Tube to Boss	3/8	-	MS51525B6	5
Tube to Boss	1/2	-	MS51502B8	5
Tube to Boss	5/8	-	MS51525B10	5
Tube to Boss	3/4	-	MS51525B12	5
Tube to Boss	1	-	MS51525B16	5
3. Nipple, Tube: Steel, 37 degree male flare and male straight thread on each end per MIL-F-18866.				
Tube	5/16	-	MS51501B5	5
Tube	3/8	-	MS51501B6	5
Tube	1/2	-	MS51502B8	5
Tube	5/8	-	MS51501B10	5
Tube	3/4	-	MS51501B12	5
Tube	1	-	MS51501B16	5

1/ These adapters may also be referred to as "fittings" and/or "couplings".

3.6.4 Hose assembling machine. A hose-to-adapter assembling machine shall be furnished to facilitate the attachment of hose adapters specified in table II in size 1/4 inch through 2 inches to the hoses specified in table I. The hose assembling machine shall be the bench mounted type capable of being fastened to the bench. The output torque at stall shall be at least 8000 in-lbs. The hose assembling machine shall operate from 115 volts, 60 hertz, single phase power source. The machine shall consist of a rotating chuck capable of gripping the hexagonal driving portion of a

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standard screw-together reusable nipple or coupling socket and a sliding self-aligning vise capable of gripping the hose and socket. The chuck shall be capable of rotating at a speed of at least 30 revolutions per minute (rpm). The hose assembling machine shall be designed so that the hose adapters are visible while being attached to the hose. The hose assembling machine rotation shall be reversible by a directional switch with a plate plainly showing whether the socket will be screwed "on" or "off" or the nipple screwed "in" or "out". A guard shall be provided to protect the operator from the moving parts. The guard shall be interlocked with the directional switch so that the guard cannot be moved from its guarding position unless the selector switch is in the "off" position. The interlock shall not allow switching from the "off" position until the moveable guard has been returned to the guarding position. The main axis of the machine shall be horizontal and the area required for accommodating the machine shall not exceed 3 feet square, exclusive of the area required in front of the machine for the operator and the hose or hose assembly being assembled or disassembled.

3.6.5 Hose cut-off machine. Each hose assembly outfit shall include a hose cut-off machine which will cut four spiral wrap hose up to 2 inches inside diameter. The hose cut-off machine shall be supplied with two hardened steel knife edge circular blades. The hose cut-off machine shall operate from 115 volt, 60 hertz, single phase power source and be in accordance with Fort Belvoir Drawing D13221E6701.

3.6.6 Hose skiver. A hose skiver for removing the outer rubber covering of the hose ends prior to the assembly of the adapters shall be provided. The hose skiver shall accommodate hoses of 1/4 inch through 2 inches inside diameter. The skiver shall be steel and equal to Caterpillar Tractor Part Number 7S1126 and shall include cutter blade equal to Caterpillar Tractor Part Number 7S1120.

3.6.6.1 Mandrels. Steel or aluminum mandrels to support the hose during skiving shall be provided in the following sizes: 1/4 inch, 3/8 inch, 1/2 inch, 3/4 inch, and 1 inch. Each mandrel shall be 15 inches long and may be heavy wall tubing per SAE J525.

3.6.7 Hose oil. No less than 1 quart of hose oil shall be provided to lubricate the hose ends and adapters prior to assembly.

3.7 Paint and protective finish. Paint and protective finish shall be in accordance with the best practice of the commercial industry. All surfaces to be painted shall, immediately prior to painting, be cleaned, free of all foreign matter, and dried. The protective paint coating shall be at least 2.0 mils (.002 inch) thick. The paint shall withstand the application and removal of a strip of fiberglass tape over the two intersecting cuts in the paint without any peeling of the paint. The surface hardness of the paint shall be between the industrial standards of H through HB pencil lead. Where no protective finish is utilized (such as machine bedways, guide surfaces, etc.), the natural finish of the material or finish obtained from heat treatment is permissible, provided the surfaces are free of scale and have a protective coating to inhibit corrosion.

3.8 Identification of product.

3.8.1 Identification marking. The assembling machine and hose cut-off machine shall be marked for identification in accordance with MIL-STD-130 and shall include the National Stock Number.

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3.8.2 Hose identification. The hose shall be marked at least every 18 inches on the layline. Order of marking may be in any sequence. The marking shall include but shall not be limited to the following:

a. R2 for Type 100R2 hose, R10 for Type 100R10 hose.

b. Nominal size 1/4, 3/8, 1/2, 3/4, 1, 1-1/4, 1-1/2, or 2 inches.

3.8.3 Adapter identification. The fittings shall be identified as specified in MIL-F-52525 and MIL-F-18866 as applicable.

3.9 Workmanship. The quality of workmanship imparted to the hose assembly outfit and all of its components shall equal or exceed that typically provided to commercial products by domestic producers of the type of item addressed herein. The items presented for acceptance shall have been manufactured with skill and care, shall be uniform, neat, and clean, and shall be free of irregularities and anomalies which degrade form, fit, function, performance, and appearance.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specifications where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that the products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1)

b. Quality conformance inspection (see 4.2.2)

4.2.1 First article inspection. When specified (see 6.2.1), first article inspection shall be performed on the preproduction model(s) or initial production item(s) noted in 3.1. First article inspection shall consist of the examination in 4.3, all tests in 4.4, and the packaging inspection in 4.5. Failure of the first article to pass the examination or any of the tests shall be cause for rejection.

4.2.2 Quality conformance inspection.

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4.2.2.1 Inspections and tests. Quality conformance inspection shall be applied to production units offered for acceptance under the contract. Quality conformance inspection shall consist of a through h as follows, and failure of any unit to pass an examination or test shall be cause for rejection of the unit.

a. Product examination (see 4.3)

b. Hose assembling machine test (see 4.4.1)

c. Hose cut-off machine test (see 4.4.2)

d. Hose skiver test (see 4.4.3)

e. Hose/adapter assembly test (see 4.4.4)

f. Salt spray test (see 4.4.5)

g. Paint adhesion test (see 4.4.6)

h. Packaging inspection (see 4.5)

4.2.2.2 Sampling. Sampling for quality conformance inspection shall be performed in accordance with MIL-STD-105, using the following:

Inspection or test	Level	AQL
Product examination (4.3)	II	1.0
Hose assembling machine test (4.4.1)	II	1.5
Hose cut-off machine test (4.4.2)	II	1.5
Hose skiver test (4.4.2)	II	1.5
Hose-adapter assembly test (4.4.4)	II	1.5
Salt spray test (4.4.5)	S-2	4.0
Paint adhesion test (4.4.6)	S-4	2.5
Packaging inspection (4.5)	S-4	4.0

4.3 Product examination. The hose assembly outfits shall be visually, manually, and dimensionally examined to determine conformance with the requirements of 3.2 through 3.4.6 and 3.6 through 3.9. Visual examinations shall include verification of completeness of manufacture and assembly, conformance to specified standards, adequacy of markings, proper cleaning, and freedom from the identified defects. Manual examinations shall include the operation of movable parts by hand to assure proper functioning. Dimensional examination shall verify size and weight conditions when specified. The examination provisions may be applied at the earliest practical point in manufacture at which it is feasible to inspect for acceptance without risk of change in the characteristic by subsequent operations. Failure of the contractor to provide objective evidence that the outfits and their components have passed the examinations prescribed for them by the contractor's inspection system shall be cause for rejection. In addition failure of the contractor to provide objective evidence that all parts are manufactured to definite standards, clearances, and tolerances so that no replacement part will degrade the form, fit, or function of the end item (see 3.4.7), shall be cause for rejection.

4.4 Tests. The following tests are applicable to the hose assembly outfits. Failure to pass any of the test requirements shall be cause for rejection of the lot.

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4.4.1 Hose assembling machine. Connect the machine to a power source of 115 volts, 60 hertz, single phase.

a. Measure the output torque at stall. The output torque shall be 8000 in-lbs minimum (see 3.6.4)

b. Verify that the chuck rotates at least 30 rpm (see 3.6.4)

Failure to meet either of these requirements shall be cause for rejection.

4.4.2 Hose cut-off machine. Connect the machine to a power source of 115 volts, 60 hertz, single phase. Cut a short length of 2 inches inside diameter, four spiral wrap hose. (This may be a length appropriate for the adapter assembly test in paragraph 4.4.4.) Failure to cut the hose cleanly, so that it can be fitted with an adapter, shall be cause for rejection (see 3.6.5).

4.4.3 Hose skiver. Remove the outer rubber covering of a length of 2 inch inside diameter four spiral wrap hose with the skiver. Failure of the outer covering to be removed cleanly, without damage to the spiral wrap, so that the hose can be fitted with an adapter, shall be cause for rejection (see 3.6.6).

4.4.4 Hose/adapter assembly. Using a short length of hose, 18 inches plus or minus 6 inches, from each of the following sizes in table I: 3/8 inch and 3/4 inch I.D., Two Wire Braid, and 1 inch and 2 inch I.D., Four Spiral Wrap, assemble with appropriate hose-to-tube adapter assembly or hose-to-flange adapter assembly. Upon completion of assembly, pressure check (with hydraulic tester) each hose assembly at 125 percent plus or minus 5 percent of the hose's rated pressure.

4.4.5 Salt spray test. Hose adapters specified in table II shall be subjected to a salt spray test in accordance with ASTM 117.

4.4.6 Paint test. The hardness, adhesion, and thickness of the paint used on the machine shall be checked as follows (see 3.7):

a. The graphite point of a sharpened HB lead pencil shall be pressed at a right angle into a painted area of the unit until the graphite crumbles. Failure of the surface to withstand the test, as evidenced by a pitting of the paint, shall be cause for rejection.

b. A test sample of the same metal, painted with the same process and at the same time as the unit, shall have a sharpened H lead pencil pressed into the paint as in a. Failure of the surface to pit (paint too hard) shall be cause for rejection.

c. Two 2 inch intersecting lines shall be cut through the paint to the base metal of the test sample. A strip of 1/2 inch wide fiberglass tape shall be firmly pressed over the intersection and then pulled from the surface. Adhesion of any paint to the tape shall be cause for rejection.

d. Using a paint thickness measuring instrument, or by masking a portion of the test sample during painting, measure the thickness of the paint. Failure of the paint to be at least 0.002 inch thick shall be cause for rejection.

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4.5 Packaging inspection. The hose assembly outfits shall be inspected before and after packaging to determine compliance with the requirements of Section 5.

5. PACKAGING

5.1 Disassembly. When practical, for the protection of components, attachments and accessories against damage, pilferage or to reduce cubage, items may be disassembled only to a necessary basic degree. Disassembled parts shall be preserved, anchored, braced, blocked, and cushioned to prevent damage. Disassembled parts shall be clearly and legibly marked as to identity and proper location on the assembled item. All fasteners removed during disassembly shall be secured in one of the mating parts. A part shall not be removed from an assembly unless it can be reassembled readily in the field without special skills or tools.

5.1.1 Disassembled parts marking. Disassembled parts shall be marked, when necessary with instructions, to facilitate reassembly. Removed parts and mating parts on the item shall be marked by stencility, on the part, or by use of tags. Tags shall be firmly secured to each individual part.

5.2 Preservation-packaging. Preservation-packaging shall be level A or level C, as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Cleaning and drying. All components shall be cleaned and dried in accordance with applicable procedures of MIL-P-116.

5.2.1.2 Preservation. Immediately after cleaning and drying, apply a coat of type P-10, grade 30 preservative oil to all bare ferrous metal surfaces of the assembly machine and hydraulic unit(s). Dies shall be preserved by the application of Type P-2 preservative. Preservatives shall be in accordance with MIL-P-116. Hydraulic unit(s) shall be filled with manufacturer's hydraulic oil, providing it has preserving capabilities or with oil conforming to MIL-H-6083. The machine shall be actuated to assure that all interior surfaces of the hydraulic system have been coated.

5.2.1.3 Unit packaging. The unit method of protection shall conform to Method I of MIL-P-116. External preserved surfaces shall be covered with barrier material conforming to MIL-B-121, Grade A. Secure barrier material with tape conforming to PPP-T-60, type optional. The assembly machine, dies, and hydraulic unit(s) shall be packaged in a close-fitting fiberboard box conforming to PPP-B-636, class weather resistant. Fiberboard conforming to PPP-F-320, class weather resistant, Type CF shall be formed into pads or cut shapes and utilized to isolate components and restrict movement. Protective cap-plugs conforming to MIL-C-5501 shall be placed over the quick coupler and fluid outlets. All vent holes shall be sealed with tape to prevent any leakage of fluids. Fittings may be packaged in multiple quantity packages, provided that thread protectors are used on the fittings.

5.2.1.3.1 Hoses and fittings. Hoses and fittings shall be unit packaged in accordance with Level A requirements of MIL-P-775. Hoses shall be coiled to minimum safe diameter and packaged accordingly. The hoses may be packaged in multiple quantity unit packages, within one unit package.

5.2.1.3.2 Technical data. Technical data shall be packaged in accordance with Method IC-1 of MIL-P-116 and placed in the unit container housing the assembling machine.

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5.2.2 Level C. Level C preservation-packaging shall be in accordance with MIL-STD-1190 to be preserved and packaged in a manner to afford adequate protection against corrosion, deterioration, and physical damage during shipment and storage.

5.3 Packing. Packing shall be Level A, B, or C, as specified (see 6.2).

5.3.1 Level A. Each hose assembly outfit shall be packed in a cleated plywood or nailed wood box conforming to PPP-B-601, overseas type or PPP-B-621, Class 2, respectively. Components shall be blocked, braced, and cushioned in accordance with MIL-STD-1186.

5.3.2 Level B. Each hose assembly outfit shall be packed in a box conforming to PPP-B-601, domestic type, Style A, B, or J, PPP-B-621, Class 1 or PPP-B-640, Class 2. Blocking, bracing, and cushioning shall be as specified in 5.3.1.

5.3.3 Level C. Each hose assembly outfit shall be packed in accordance with MIL-STD-1190 in such a manner to afford adequate protection against physical and mechanical damage during shipment and storage.

5.4 Marking. Marking, for all levels of interior packages and exterior shipping containers shall be in accordance with MIL-STD-129. Any special markings, if needed, shall be required by the contract. Bar Code marking is also required and shall be in accordance with MIL-STD-129 and MIL-STD-1189.

6. NOTES

(This section contains information of a general explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The hose assembly outfit covered by this specification is intended for use in the fabrication and repair or replacement of hydraulic hose assemblies.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. First article when required (see 3.1).

c. First article inspection if different (see 4.2.1).

d. Level of preservation-packaging and packing, required (see 5.2 and 5.3.)

6.3 First article. When a first article inspection is required, the hose assembly outfit should be a first article sample, a first production item, a sample selected from the first run of production items or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The first article should consist of a specific number of units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering products which has been previously acquired or test by the Government, and

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that bidders offering such products, who wish to reply on such production or test, must furnish evidence with the bid that prior Government is presently appropriate for the pending contract.

6.4 Data requirements. Required technical data such as operator's manuals, parts lists, and other instructions for operation and maintenance, as identified on a numbered DD Form 1664, should be specified on DD Form 1423 incorporated in the contract.

6.5 Manuals.

6.5.1 Instruction manual. An operator's instruction manual which details all phases of repair and fabrication of hose assemblies should be provided. As a minimum, it should provide the following information:

a. Adapters disassembly from and assembly to hose, and required torque.

b. Preparation of hose ends: cutting to length, skiving and application of lubricating oil.

6.5.2 Repair part manuals. Repair parts and accessory identification in repair part manuals should include the prime contractor's name, nomenclature, and part number. Repair part identification in repair part manuals should also include the Original Equipment/part Manufacturer's (OEM) (actual part manufacturer's) name, Commercial and Government Entity (CAGE), nomenclature and part number of the part used in the hose assembly outfit, immediately after the prime contractor's part identification. Items with high mortality rates should be so identified.

6.6 Safety and health requirements. In order that equipment integrated into the user's operational environment will comply with OSHA limitations and control of noise levels, radiation, electronmagnetic emission, noxious vapors, heat, etc., as applicable, specific requirements concerning such points of operation and other health and safety requirements, should be specified by the user.

6.7 Subject term (key word) listing.

Hose Assembly Outfit
Castings
Fastening Devices
Finish
Forging
Interchangeability
Safety
Threads
Welding
Workmanship

Custodian:
Army - AL

Preparing activity:
Army - AL

Review activities:

Project No. 4940-A571

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

MIL-H-65054(AL), Hose Assembly Outfit (Reusable Adapters)

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 68 WHICH MAY BE USED

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